

Secure Hybrid Datacenter Network Architecture & Multi-Tier Cloud Segmentation

Zero Trust • AWS Multi-VPC • IAM • Kubernetes • STRIDE • Automated Validation

Raj Chittoriya

Roll No. 2410030901 • Section 3CSE15
B.Tech CSE (Cyber Security) • IILM University, Greater Noida

CISCO Virtual Internship Program 2026

Domain: Ethical Hacking
Duration: March 2026 – June 2026

Submitted for Internship Evaluation

Internship Overview

Program

CISCO Virtual Internship Program 2026

Domain

Ethical Hacking

Duration

March 2026 – June 2026

Mentor

Mr. Abhinav Raghav
Cisco Networking Academy, IILM University

Learning path

Ethical Hacking

Security Fundament

Social Engineerin

Malware

Network Essentials

Kali Linux

Basic Web VA

Basic Web PT

Capstone Project

Assessmen t

Internship → practical capstone progression

Foundation

Attack concepts, security fundamentals and VAPT.

Architecture

Zero Trust + hybrid cloud segmentation.

Validation

Automated security-policy verification.

Internship & Project Objectives

01

Ethical Hacking

Build attacker-oriented security understanding.

02

VAPT

Understand vulnerability assessment and penetration testing.

03

Networking

Learn segmentation, routing and secure architecture.

04

Cloud Security

Apply VPC, SG, NACL and Transit Gateway concepts.

05

Identity

Apply least privilege, MFA and STS.

06

Kubernetes

Use namespaces, RBAC and NetworkPolicies.

07

Threat Modeling

Map threats using STRIDE.

08

Automation

Validate policies with Python.

Problem Statement

A hybrid environment creates more trust boundaries. The architecture must stop a single compromise from becoming unrestricted lateral movement.

Hybrid connectivity

Cloud and private datacenter must communicate securely.

TGW + dual IPsec VPN

Workload isolation

Student, teaching and research workloads have different trust levels.

Separate VPCs / namespaces

Database exposure

Private databases cannot be reachable directly from the internet.

Private DB + SG/NACL

Lateral movement

A compromised application must have a small blast radius.

Segmentation + NetworkPolicy

Identity abuse

Privileges must be limited and admin escalation blocked.

Least privilege + MFA + Explicit Deny

Visibility

Security events need centralized evidence and auditability.

CloudTrail + Flow Logs + SIEM

Technologies & Tools

AWS

VPC • Security Groups • NACLs • Transit Gateway • IAM • WAF • CloudTrail • VPC Flow Logs

Kubernetes

Namespaces • RBAC • NetworkPolicies • Pod Security Context

Cisco ASA

Modelled stateful firewall at datacenter edge

IPsec VPN

IKEv2 • AES-256-GCM • SHA-384 • DH Group 20

TLS 1.3

Encryption in transit for public-facing traffic

Kali Linux

Security testing / penetration-testing environment

Python 3.10

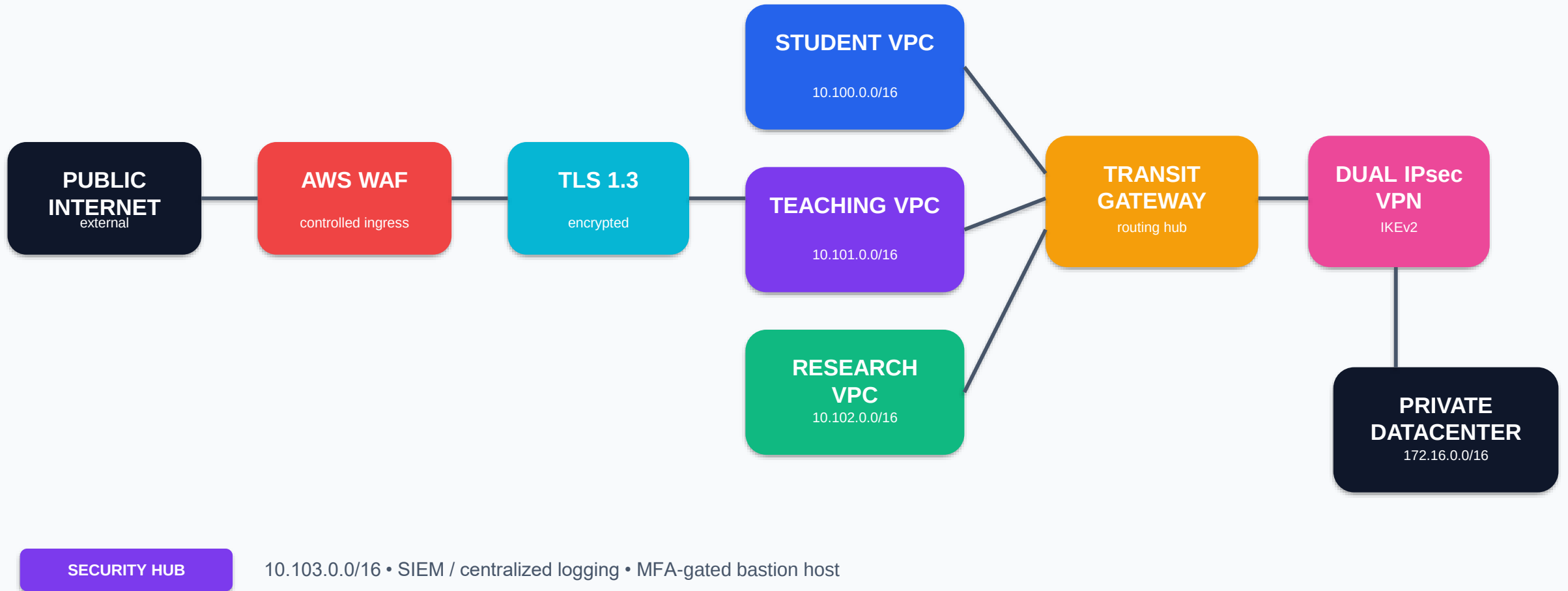
Automated security-policy verification engine

STRIDE

Structured threat modelling framework

High-Level Hybrid Architecture

Controlled ingress → segmented cloud → encrypted hybrid connectivity → protected private datacenter



VPC Segmentation & IP Addressing

Network	CIDR	Subnets / Components
Student VPC	10.100.0.0/16	Public ALB 10.100.1.0/24 • App 10.100.10.0/24 • DB 10.100.20.0/24
Teaching VPC	10.101.0.0/16	Public ALB 10.101.1.0/24 • App 10.101.10.0/24 • DB 10.101.20.0/24
Research VPC	10.102.0.0/16	HPC Compute 10.102.10.0/24 • Data Vault 10.102.20.0/24
Security Hub	10.103.0.0/16	SIEM / centralized logging • MFA-gated bastion
Private Datacenter	172.16.0.0/16	LDAP/AD 172.16.10.10 • Enterprise Core DB/ERP 172.16.20.50

Key design rule: non-overlapping CIDRs make segmentation and routing boundaries explicit.

Layered Security Controls

Edge

AWS WAF • TLS 1.3 • rate limiting • controlled ingress

Network

Non-overlapping VPCs • isolated subnets • SGs • NACLs • TGW route controls

Database

Private backend-only access • no direct internet exposure

Identity

Least privilege • MFA • STS • SecureTransport • Explicit Deny

Kubernetes

Namespace isolation • RBAC • default-deny NetworkPolicy • pod hardening

Monitoring

VPC Flow Logs • CloudTrail • centralized SIEM/log aggregation

Datacenter

Cisco ASA stateful firewall • controlled LDAP • protected ERP/Core DB

IAM + Kubernetes Security

Role-StudentWorkload

Read-only student asset access; administrative actions explicitly denied.

Role-Faculty

Course content access; deletion requires hardware MFA.

Role-SecOps-Auditor

Read-only CloudTrail + VPC Flow Logs; log deletion forbidden.

Kubernetes security model

student-workloads

teaching-workloads

research-workloads

security-monitoring

Default-deny

Only explicitly authorized pod communication is allowed.

RBAC

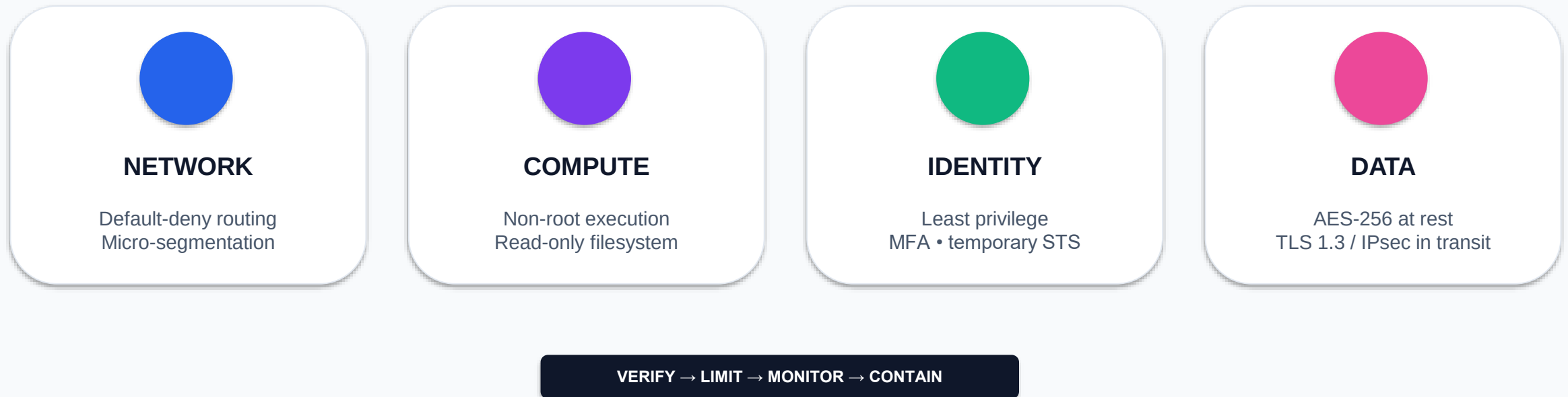
Kubernetes API permissions are limited by role.

Pod hardening

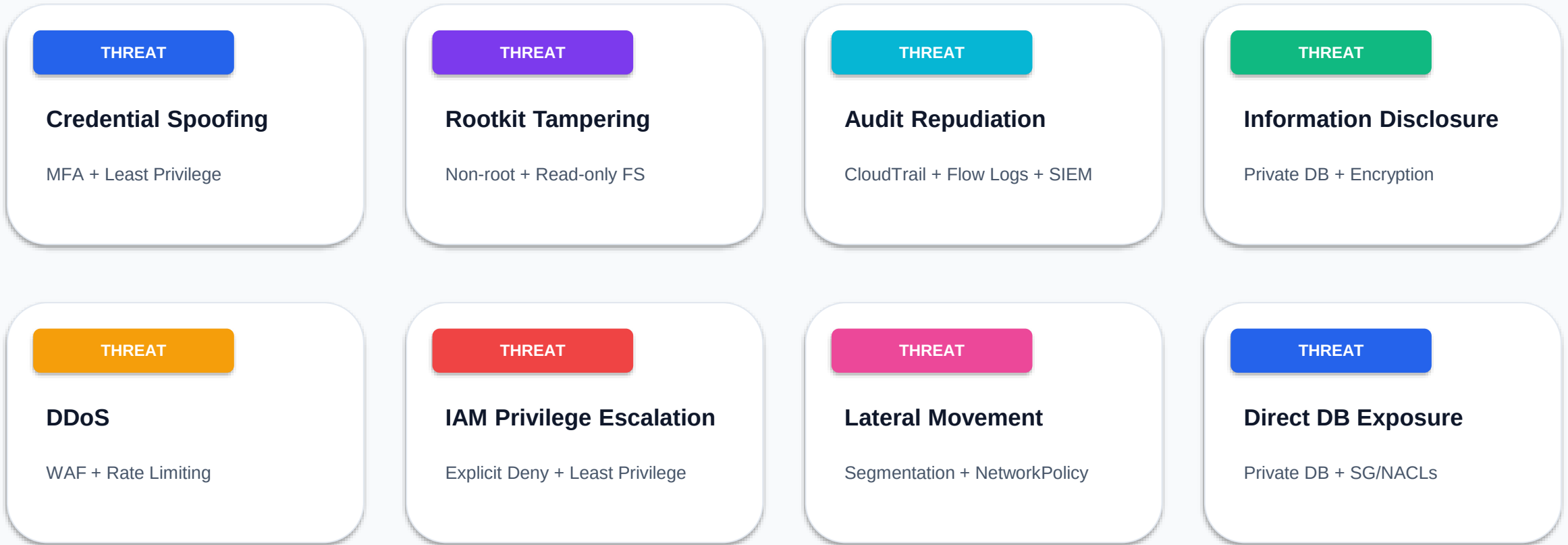
runAsNonRoot • readOnlyRootFilesystem •
token automount=false

Zero Trust Security Model

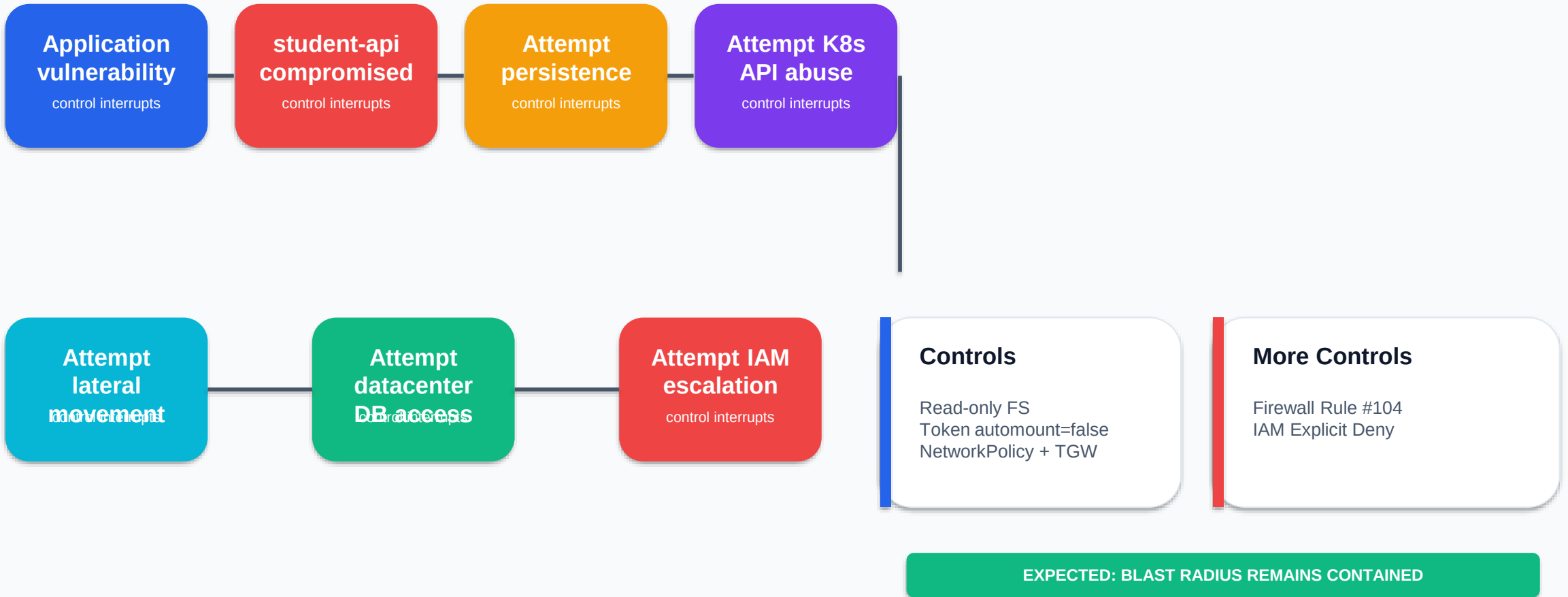
“Never trust automatically; verify explicitly and enforce least privilege.”



STRIDE Threat Model



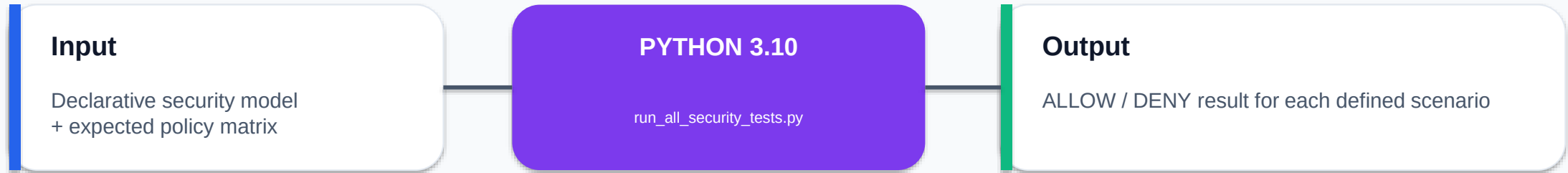
Critical Attack Scenario & Containment



This is a defined security simulation/validation exercise — not a live production attack.

Automated Security Verification

Python 3.10 engine replaces manual case-by-case inspection with repeatable policy validation.



Core validation categories

Network boundaries

Database exposure

Kubernetes namespaces

IAM escalation

Command

```
python tests/run_all_security_tests.py
```

Purpose: validate that defined ALLOW/DENY behavior matches policy intent.

10-Point Security Test Matrix

1	Internet → Public HTTPS ALB :443	ALLOW
2	Internet → Private DB :5432/3306	DENY
3	Frontend/Web → Backend API	ALLOW
4	Frontend/Web → Private DB Direct	DENY
5	Backend API → Database	ALLOW
6	Student VPC → Research VPC	DENY
7	Student VPC → Datacenter Core	DENY
8	Unauthorized Cross-Namespace	DENY
9	Authorized Intra-Namespace	ALLOW
10	Unauthorized IAM/Admin Escalation	DENY

10 / 10
TESTS PASSED

100%
COMPLIANCE
WITH DEFINED MATRIX

Meaning: the declarative security model matched the expected ALLOW/DENY outcomes.

Individual Contribution & Scope

Role

Project Lead & Individual Developer

Key responsibilities

Hybrid architecture + VPC segmentation

SG/NACL + IAM/MFA policy modelling

Kubernetes namespaces + RBAC + NetworkPolicy

IPsec/TGW + STRIDE threat modelling

Python verification engine + 10 tests

README + report + presentation + evidence

Locally Implemented / Modelled

VPC configurations • SG/IAM definitions • Kubernetes manifests • firewall definitions • Python validation • evidence logs

Simulated / Conceptual

Physical Cisco hardware • Live AWS VPC/TGW • live EKS • physical enterprise datacenter

Accurate wording: designed • modelled • simulated • validated

No claim of a live production AWS penetration test is made in the report.

Learning Outcomes, Certificates & Future Scope

What I learned

Zero Trust • VAPT • network segmentation • cloud security • IAM • Kubernetes • threat modelling • automation

Future Scope

Trivy + Checkov • Cilium + Falco • Direct Connect + MACsec • continuous regression testing

Certificates

Cybersecurity Essentials
20 Jun 2026
ID: cb63b578-677e-48ec-8a42-fb5e7c87b7c9

Getting Started with Cisco Packet Tracer
16 Jun 2026
ID: 2dbfb49c-4cdb-4563-92f0-272a202b0839

Certificate

Exploring Networking with Cisco Packet Tracer
16 Jun 2026
ID: 6818ce3e-d501-4163-b212-915bd83d4789

Issuer / Provider: Cisco Networking Academy, IILM University

Closing Takeaway

Security should be layered and explicit.

The project demonstrates how Zero Trust can contain compromise through segmentation, least privilege, encryption, monitoring and automated validation.

THANK YOU • QUESTIONS?

Q & A

“Never trust automatically; verify explicitly and enforce least privilege.”

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CISCO - AICTE Virtual Internship 2026 • Ethical Hacking

SECURE • SEGMENT • VERIFY